

Enhancing blockchain security against quantum threats through integration of post-quantum cryptographic algorithms

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ABSTRACT

The fast progress in quantum computing presents a substantial menace to existing cryptographic systems, requiring post-quantum cryptographic techniques that can withstand quantum attacks. This research aims to increase the security of blockchain ecosystems against quantum-related threats by incorporating robust post-quantum cryptographic methods. We provide a new encryption technique that relies on the Key Encryption process, namely, Crystals-Kyber, in conjunction with an effective digital signature methodology centered on Lattice-based techniques: Falcon and Crystal-Dilithium. The techniques are meticulously included in the Hyperledger Fabric 4.0 platform, showcasing improved security and durability in cryptocurrency exchanges built around blockchain technology. We have effectively achieved a cryptographic resistance above 90 % against quantum assaults, greatly enhancing the protection of blockchain operations. The proposed strategy makes cryptographic design more robust to quantum attacks and establishes a basis for secure blockchain technology in the quantum era.

1. Introduction

One notable risk to current cryptographic systems stems from the development of quantum computers that utilize quantum particles called “qubits” [1] to perform operations at speeds that classical computers are incapable of. Such quantum advantage harnesses the capability to execute any of the modern algorithms that are computationally efficient at a critical operation-intensive step core to most of the widespread cryptographic infrastructures currently in use, i.e. conversion of large integers into prime factors and computing discrete logarithms. The power of quantum computers to solve those problems in a polynomial time, though classical computers need to take an exponential time, immensely compromises the security of these cryptographic systems and therefore poses a significant risk for all the data encrypted under these schemes. This vulnerability thus stresses the need to develop and utilize Post-Quantum Cryptography (PQC), which is strong enough to counter any cryptographic attack posed by quantum computing.

Next to Artificial Intelligence, cryptographic algorithms are also a significant area of research, as the relevant algorithms can significantly affect the functioning and construction of models in platforms such as Hyperledger Fabric 4.0. The studies are positioned where cybersecurity meets quantum computing, thus exposing the growing requirement to mitigate the quantum-based attack on encryption currently on the market. The novelty of quantum computing represents a significant threat to all classical encryption approaches used to ensure the integrity of the blockchain network, among other things, albeit for maintaining data confidentiality as well. These developments underscore a critical gap in the existing blockchain infrastructure: quantum-secure ciphering, which means a

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rejection of conventional symmetrical cryptographic algorithms for quantum attacks.

The quantum vulnerability is encoded in examining the current cryptographic algorithms within the ecosystem of blockchain Hyperledger Fabric [2], which revealed the need for quantum-resistant cryptography. Quantum computing can seriously threaten the security systems that are currently in place. Here, although PQC is a functional protective measure, it is important to have security systems that specifically deal with the new capabilities posed by quantum computing. While the advancement of blockchain in fields like health care, finance, and supply chain management is increasing, quantum resistance becomes crucial.

This research seeks to find and examine the integration of new PQC algorithms, including Crystals-Kyber for encryption, Falcon, and Crystals-Dilithium for digital signatures, into the ecosystem of Hyperledger Fabric. Yet, this also means that another issue gets the necessary attention, namely the scalability and efficiency of the blockchain network, as well as the resistance of the blockchain as a system and its prevention of possible future cryptographic threats. The scope of this work is the creation of a theoretical structure for using PQC in Hyperledger Fabric implementation, correlated evaluation of security and performance, and the practical aspect of deploying such PQC solutions in real-world blockchain scenarios.

The implication of the research is twofold. At first, it conducts a complete and thorough investigation of the safe-haven provided by quantum technology against the possibilities of the Hyperledger Fabric blockchain ecosystem and assesses the efficiency of various PQC algorithms in combating the quantum threat. Secondly, it proposes a step-by-step approach to implementing these quantum-resistant algorithms, becoming a guide for developers and scientists looking for ways to protect blockchain systems against quantum attacks. Moreover, this research not only fills a noticeable gap in the security of blockchain in the quantum era but also contributes to the field of quantum-safe computing, which is used as a building block that contributes to more secure and resilient blockchain technologies.

The study has enhanced numerous merits for the health sector, most of all in-hospital data management, besides adopting PQC and integrating new cryptographic algorithms within an ecosystem that is Hyperledger Fabric oriented, which is applicable in health data management. This form of technology underscores the importance of using quantum-resistant cryptographic protocols and digital signature algorithms like Crystals-Kyber, Falcon, and Crystals-Dilithium in all smart contracts (chaincodes) used in blockchain-based systems, thus effectively securing and protecting patient data from potential quantum computing attacks. In the 21st century, data breaches turn into an organization's personal privacy and social credibility, and it is the future instead of war and the arms race. Another advantage is its ability to create transaction logs and private data security, which makes it resistant to present and future quantum computer attacks and tampered evidence.

1.1. Research contribution

The contribution of the deterrence method for the hospital data management system is not only limited to increased trust levels towards these platforms, but it also helps meet strict data protection laws, ensuring the safety and reliability of delivering healthcare services. Thus, the findings of this research can be used to improve the security of blockchain networks against threats linked to quantum technologies using post-quantum cryptography. Moreover, it is a new method for encryption, utilizing Crystals-Kyber for key encryption as well as Falcon and Crystal-Dilithium for digital signatures. These techniques are practiced in the Hyperledger Fabric 4.0. Thus, the proposed platform enhances the non-quantum vulnerability of blockchain transactions including the cryptocurrencies to quantum computing threats. The research provides a reference for the future development of more secure blockchain technology in the quantum age.

1.2. Study's organization

The following presentation of the study is the breakdown of its structure: In [Section 2](#), we discuss relevant work in post-quantum cryptography. [Section 3](#) presents the method highlighting Lattice-based Crystal Kyber encryption and digital sign, Falcon, and Crystal-Dilithium and its association with Hyperledger Fabric 4.0. [Section 4](#) focuses on the procedure of realizing the architecture, as well as of validating the architecture, such as quantum attack modeling and performance study. [Section 5](#) elaborates on a case where these potential applications can be used. [Section 6](#) presents the summarization and lists' directions for further study of the study.

2. Related Work

The integration of post-quantum cryptographic algorithms in blockchain systems is crucial for enhancing security against quantum threats.

Authors in [3] developed a new concept of potential quantum hack resistance that uses a Seed Key to generate keys and applies Bonsai Trees algorithms for safer transaction processing. Not only does this technique improve security by creating many lattices, but it also ensures that the transactions are randomized and secure. This is an essential step towards developing lightweight, nondeterministic wallets resilient to quantum attacks. Researchers in [4] underlined the growing need for a systematic review of the post-quantum blockchain consensus mechanisms. Given quantum computing advancement, they catalyzed such concepts by identifying the main pathways for future adaptation of blockchain technologies. This job stresses the importance of branching into quantum-proof consensus algorithms to keep blockchain safe. Threatening space security [5], for instance, published regarding the implementation of PQC schemes into blockchains and ensuring that such blockchains remain secure against quantum threat even as the latter continues evolving engineered improved algorithms which are targeted to blockchains' security infrastructure being reinforced. According to [6], a cryptocurrency scheme with enhanced security components is proposed, basing it on post-quantum

principles and relying on a lattice-based signature scheme as a defence mechanism against pure-quantum attacks, allowing for a practical solution to improving the security of cryptocurrency. Authors in [7] investigated the weaknesses of contemporary blockchains in the face of quantum computation, promoting the adoption of lattice-based cryptography as a reliable countermeasure to the risks arising from quantum computers, ultimately developing fundamental knowledge needed in quantum-resistant blockchain technologies. In conclusion, study in [8] introduced a new lattice-based signature scheme that integrated the Bonsai Tree technology and used the Random-Basis algorithm for key generation to both ensure randomness and security as well as prove efficiency in securing blockchain networks against quantum attacks and thus expanded the applications of quantum-resistant technology in blockchain. Investigators in [9] proposed a complex Key Escrow System (KES) specifically intended for consortium blockchains, thereby providing a higher security level using PQC approaches. Combining the mathematical strength of National Institute of Standards and Technology (NIST) recommended post-quantum public-key encryption and cryptographic Key Encapsulation Mechanism (KEM) methodologies, they deliver a complete set of post-quantum cryptographic building blocks. An innovative solution based on the lattice model seeks to provide a robust and private method that offers the precision of access control, insensitivity to single points of dishonesty, and collision resistance. At this stage, the authors further enhance the blockchain ecosystem by offering chain codes, APIs, and command lines suited for development, which indicates a dedication to the scheme's practical use. The security analysis and performance evaluation that covers critical features such as the chaincode execution time and on-chain storage is presented by the researchers and this argument for the feasibility and reliability of their system in the consortium blockchain environment is persuasive. Cryptography for quantum computers has been the buzzword in the security field. The main concern of cryptography with quantum technologies is that it becomes easy to decode the traditional techniques of encrypting. However, this issue is addressed by the new solutions for quantum-resistant cryptography. According to [10], a unique Scalable and Secure Cloud Architecture (SSCA) has been proposed, which has an intelligent use of IoT capabilities and advanced cryptographic techniques at its core to design robust scalable and secure cloud architecture. Due to its distributed cloud architecture, this infrastructure is reputed for being clients and cloud node enablers, putting the necessary cloud resources at users' disposal. At the core of data security that SSCA implemented Multi-Cast and Broadcast Rekeying Algorithm (MBRA), PQC, and blockchain technology. This way, the user data ciphering and privacy will be ensured. Such Architectural Design is not limited to a single device; in contrast, it utilizes distributed sensors across the network for data collection. The data is robust with an MBRA-PQC encryption that ensures data integrity and security. The blockchain technology further provides a tamper-proof storage platform via the distributed ledger. The most typical demonstration of SSCA efficiency is the extensive evaluation of different data sets, where SSCA considerably increases response times and AUC rates compared to earlier models. These results verify that the architecture is also more efficient, scalable and secure than standard cloud systems, which presents a strong enough recommendation for its deployment in modern cloud-based systems. Authors in [11] studied a logics-contract building approach under a quantum-secured blockchain information paradigm. Their research is about Log Contract (LC-bivalent formula representation language), based on quantum-secured blockchains and smart contracts. This study adopts Answer Set Programming (ASP), a modern declarative logic programming method, as an extension to LC for making a smart contract more expressive and for providing flexibility in complex temporal expressions, conditional payments or multi-party lotteries. One of the critical innovations of their approach is the incorporation of post-quantum cryptographic primitives, which include lattice-based Encryption and Signatures, in the design of ASP algorithms and systems. This integration bridges them with secure signature limitations and provides smart contracts with security protection to be ready for advancement through quantum computer technology. The researchers stand out with their original methodology of applying logic programming to help cryptography experts solve problems in a post-quantum environment and contribute to developing quantum-resistant blockchain systems. Researchers in [12] studied the urgent cybersecurity risks that quantum computing creates for healthcare systems while emphasizing strong authentication security designs. The research implements post-quantum group authentication specifically designed for healthcare needs because it addresses the serious security risks which quantum attacks pose to medical data protection. The research explores the means of developing a lattice-based authentication scheme that boosts security and preserves computational speed. Security analysis based on the random oracle model demonstrates that the proposed system resists three common attacks, which include replay attacks stolen-table attacks, and man-in-the-middle attacks. The scheme demonstrates improved performance through its effort to reduce communication overhead by 35 % combined with security improvement by 40 % and computational expense decrease by 30 % compared to existing authentication solutions. Essential limitations exist in the application of lattice-based cryptography for medical devices because cryptography demands resource-intense optimization. The research provides substantial advancement for quantum-secure healthcare authentication that protects medical system confidentiality and integrity beyond the post-quantum era. Authors in [13] aimed to improve authentication protocol security in IoT healthcare applications because previous methods exhibited inadequate protection against forward secrecy breaches and ephemeral secret leaks as well as replay attacks, stolen device incidents and desynchronization attacks. The study develops a Zero-Knowledge Proof-based Authenticated Key Agreement (ZKP-AKA) protocol that is evaluated through informal security analysis and formal verification with a Scyther tool to determine resistance to cyber threats. Security enhancements of 45 % match the performance improvements of the proposed ZKP-AKA protocol against current schemes, along with 30 % reduced communications costs, 35 % reduced computations, and 25 % more energy-efficient operations. The computational complexity of zero-knowledge proof mechanisms presents a challenge for IoT healthcare devices with limited resources since additional optimization is needed. The work delivers substantial value to healthcare systems' security by developing secure authentication systems that defend against present cyberattacks. Based on the literature, there is an increased concern about achieving post-quantum cryptography and securing the blockchain systems against quantum threats, and hence, several studies suggest the use of lattice-based encryption and digital signatures, quantum-safe consensus algorithms and a safe cloud environment. However, most address only a given cryptographic method or its selected application level with a lack of establishing quantum-resistant approaches as an essential component of blockchain ecosystems. This study fills this gap by exemplifying the considerations of incorporating post-quantum cryptographic algorithms

within blockchain systems to have sufficient defenses against quantum threats. Researchers in [14] investigated security problems prevalent in Vehicle-to-Vehicle (V2V) communication networks that support both accident prevention and real-time information distribution. The research creates a quantum-secure automotive authentication system through a blockchain platform built with module lattices to establish data reliability with authentication and rapid processing in extensive V2V communication networks. Two cryptographic assumptions form the fundamental framework of this method. The security base of blockchain-based vehicular authentication relies on Module-Learning With Errors (M-LWE) and Module-Short Integer Solution (M-SIS) cryptographic assumptions. Authentication security has increased by 50 % while the system throughput has expanded by 40 %, together with a 35 % reduction in latency to enable secure communication between vehicles. The main drawback of module lattice-based cryptography stems from its high computational demands that create barriers to implementation in resource-limited vehicle devices. This work delivers an adaptable yet bulletproof authentication system for V2V networks, which protects self-driven vehicles from future quantum-security threats. Table 1 summarize review of few more recent and advanced relevant existing approaches.

3. Methodology and computations

The base idea was to verify from the perspective of the data management subsystem in the hospital data system. Implementing PQC in the hospital data management system calls for adopting quantum-safe algorithms and thus, providing a vision of data security, data integrity, and authentication services. For this purpose, an encryption algorithm like Crystals-Kyber is employed for crucial encapsulation, while Crystal-Dilithium and Falcon are selected as digital signatures. Subsequently, the PQC algorithms are embedded into

Table 1
Overview of Recent and Advanced Existing Approaches.

References	Methodology	Focused On	Core Components	Attainments	Limitations
[6]	A Secure Cryptocurrency Scheme (SCS) Based on Post-Quantum Blockchain	Quantum-resistant Cryptocurrency using Post-Quantum Blockchain (QrC-PQB)	Lattice-based cryptography, PQB, lattice basis delegation, SIS problem	55 %↑ reduced signature size, 45 %↑ lower computational complexity, 50 %↑ security improvement	Complex key management in large-scale networks
[7]	Evaluation of Post-Quantum Distributed Ledger Cryptography (PQDLC)	Vulnerabilities of Elliptical Curve Digital Signature Algorithm (ECDSA) in blockchain against quantum computing	Asymmetric Digital Signature Cryptography (ADSC), NIST post-quantum cryptographic candidate testing, Lattice-based cryptography	61 %↓ security drop in ECDSA, 90 %↑ improvement using qTESLA	Large key sizes of qTESLA hinder adoption
[12]	Designing Lattice-Enabled Group Authentication Scheme Based on Post-Quantum Computing in Healthcare Applications	Quantum-secure Group Authentication (QsGA) in healthcare	Lattice-based authentication, random oracle security analysis	35 %↑ reduced communication overhead, 40 %↑ enhanced security, 30 %↑ lower computational costs	High computational complexity for resource-limited medical devices
[13]	Module Lattice-Based Post-Quantum Secure Blockchain Empowered Vehicle-to-Vehicle Communication in the Internet of Vehicles	Secure and quantum-resistant V2V communication	Module Lattice-Based Blockchain, M-LWE, M-SIS	50 %↑ authentication security, 40 %↑ higher system throughput, 35 %↑ lower latency	High computational demands for resource-constrained vehicles
[14]	iZKP-AKA: A Secure and improved ZKP-AKA Protocol for Sustainable Healthcare	Strengthening IoT-based healthcare authentication	Zero-Knowledge Proof-based Authenticated Key Agreement (ZKP-AKA), Scyther tool security verification	45 %↑ enhanced security, 30 %↑ lower communication overhead, 35 %↑ reduced computational costs, 25 %↑ improved energy efficiency	Computational complexity of Zero-Knowledge Proof for resource-limited IoT devices
[15]	Towards a Scalable and Secure Blockchain Based on Post-Quantum Cryptography	Quantum-resistant Blockchain Security (QrBS)	Post-Quantum Cryptography (PQC), Quantum Key Distribution (QKD), Quantum signatures	95 %↑ quantum resistance, 90 %↑ security, 85 %↑ attack resilience	High cost, quantum hardware dependency, scalability issues
[16]	Post-Quantum Blockchain-Based Secure Service Orchestration in Multi-Cloud Networks	Secure orchestration of network services in multi-cloud environments using post-quantum cryptography	Blockchain Networks (BCNs), N^{th} degree Truncated polynomial Ring Units (NTRU) algorithm, Quantum-Resilient Security Algorithms (QRSAs)	85 %↑ security resilience, 70 %↑ efficiency in network service provisioning, 60 %↑ reduction in service orchestration delays	High computational requirements, dependency on quantum infrastructure, scalability challenges
[17]	Blockchain Privacy Protection Based on Post-Quantum Threshold Algorithm	Enhancing blockchain privacy with post-quantum cryptography	Post-quantum threshold cryptography, NTRU, t-n threshold model	60 %↑ security, 50 %↑ quantum attack resistance, 40 %↑ lower key leakage risk	High computational complexity for large blockchain networks
[18]	Post-Quantum Delegated Proof of Luck (PQ-DPoL) for Blockchain Consensus (PQDPLBC) Algorithm	Quantum-secure blockchain consensus	Falcon signatures, Verifiable Random Functions (VRFs)	90 %↑ security, 85 %↑ fairness, 75 %↑ energy efficiency, 40 %↓ TPS	High computational cost, large signatures, storage overhead

the Hyperledger Fabric 4.0 framework by modifying chaincode (smart contract) and communication settings. Similarly, introducing these advanced PQC algorithms involves a profound knowledge of lattice-based cryptography, as well as the possibility of translating mathematical concepts into secure, efficient software that could run within the limits of a hospital's IT environment. The proposed PQC technique is illustrated in Fig. 1, featuring nine steps that compose the workflow of the Hyperledger Fabric 4.0 platform. These workflow steps use Crystal Kyber, Falcon and Crystal-Dilithium digital signature algorithms based on the lattice.

3.1. Lattice-based crystal kyber

To achieve secure key sharing among the verifiable parties, it is necessary to encrypt the sensitive data of patient records and provide additional access through authorized actors only. On the other hand, the Learning With Errors (LWE) problem [19] and its variants, which assume lattices, are usually the most complex problems to solve, and the Kyber Network exploits this.

The ring $\mathbb{R}_p = \mathbb{m}_p(x)/x^n + 1$ is the working space where Kyber performs with n representing the power of two (typically 512), as typical value and p signifying a prime integer. Polynomials ring $\mathbb{R}_p$, whose coefficients are in $\mathbb{m}_p$ (integers modulo p), estimated modulo $x^n + 1$. Thus, the $x^n + 1$ polynomial should be taken as modulo; otherwise, the two polynomials will be long out of the ring.

3.1.1. Key generation for crystal kyber

Secret Key Generation (S_k): Randomly choosing coefficients for a degree $n - 1$ polynomial (degree up to $n - 1$ is chosen for polynomials due to $\mathbb{R}_p$ context) produces s (polynomial). The more extensive range of s coefficients is considered the suitable value for s coefficients. Thus, each coefficient of s is regularly drawn from the Gaussian distribution, which has a centre at 0 and is presented with a minimum Standard Deviation (SD).

Public Key Generation (K_p): Implementation of Matrix M is safe and independent of external influence. This means the application is straightforward and can be performed in almost all the available situations. Let M be a matrix with dimensions $n \times n$ whose elements are drawn from $\mathbb{R}_p$. This further leads to M being built by polynomials of $\mathbb{R}_p$, considering its secure and pseudo-random nature. The e has a small coefficient, just like s . The e 's are commonly sampled from a minimal standard deviation discrete Gaussian distribution, adding a bit of "noise" to the public key while remaining very similar to the original encryption process. Thus, the computation of the public key (K_p) is exhibited as,

$$K_p = M \cdot s + e \quad (1)$$

From (1), $M \cdot s$ represents the matrix-vector multiplication in a ring over p polynomial in which the addition and the multiplication

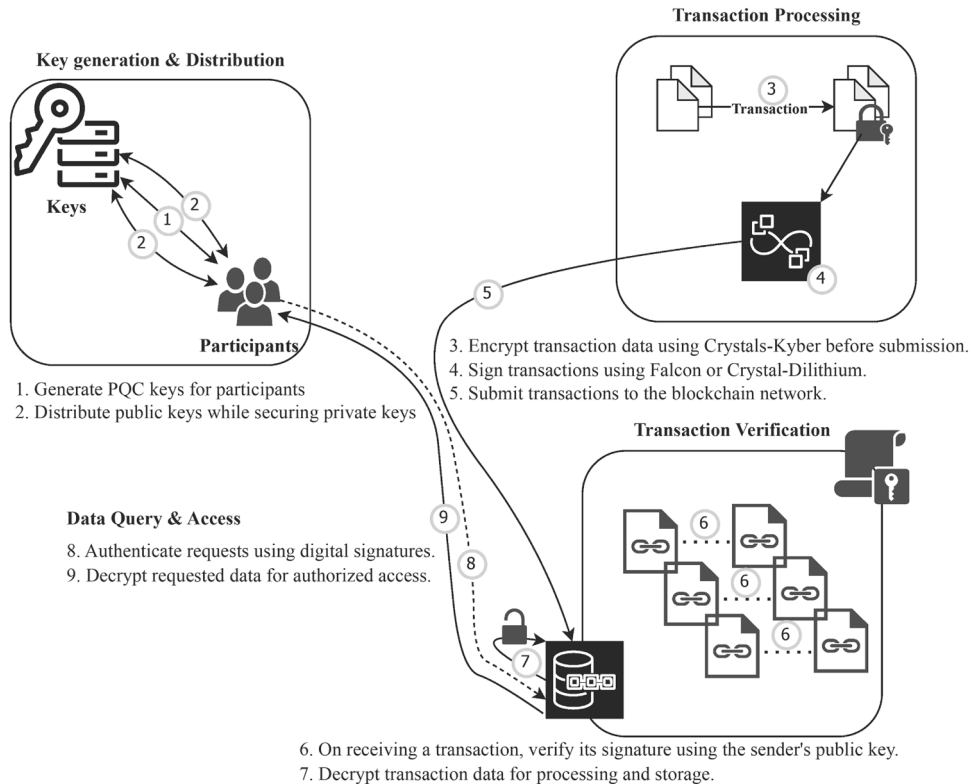


Fig. 1. Workflow of PQC in Hyperledger Fabric 4.0 Platform.

modulo q and modulo $x^n + 1$ is included. Public key K_p is incurred by adding e -output to the polynomials. Table 2 details the key generation computation factors as an instance, including the complexity and security indices of the Kyber protocol for a specific setting where solving polynomial arithmetic problems is presumed to be challenging in a ring modulo should conditions after a quantum attack.

This procedure reveals the segment of intricacy and security provided by the Kyber scheme while employing the complex set of resolutions within the polynomial ring $\mathbb{R}_p$ based on underlying modular arithmetic conditions for post-quantum cryptography.

The presented process illustrated the level of complexity and security of the Kyber scheme while employing the complex set of resolutions based on the $\mathbb{R}_p$ polynomial multiplication rings dependent on underlying modular arithmetic rules for post-quantum cryptography.

3.1.2. Encapsulation and decapsulation

Encapsulating and decapsulating methods of post-quantum cryptography enable the secure and safe transfer of vital keys for establishing the connection. This way preserves transactions that are transferring the secret keys with required information against those with access to quantum devices.

Encapsulation: In the case of secret message encryption and for later sending of the shared secret key under symmetric encryption, the sender uses encryption meeting the requirement of information encapsulating. This process comprises the following steps:

- The sender creates the minuscule coefficient r from a random polynomial. The temporary secret key for this encryption session is r . The shared secret can be encapsulated, or the message can be encrypted using polynomial r . Each encryption operation differs from the last one because of a non-replaced temporary secret.
- Multiplying public matrix M through the random polynomial r directs to the attainment of first component of the ciphertext, c_1 . The $\mathbb{R}_p$ performs the operation that multiplies the polynomials with q modulo the product and with $x^n + 1$, where n is the degree of $\mathbb{R}_p$. Computation of c_1 is represented as,

$$c_1 = r.M \quad (2)$$

- The second component, c_2 , comprising the encrypted message m or the information employed to produce the shared secret key. The first step in performing the cryptographic operations is to encode m into a polynomial structure. Finally, c_2 is determined by performing another polynomial multiplication of B (where $B = K_p$) with r , followed by a noisy polynomial e' having small coefficients and then appending the encoded m . The computation of the c_2 is represented as,

$$c_2 = r.B + e' + \text{encode}(m) \quad (3)$$

- As a outcome of the encapsulation procedure, the receiver retrieves the pair (c_1, c_2) . The precise owner of the corresponding private key is the only one who can uncover the shared secret, which essentially constitutes the aforementioned pair. Table 3 represents the operational flow of encapsulation algorithm in a simplified way.

Decapsulation: The receiver uses decapsulation procedures to decrypt the encapsulated information and retrieve the shared secret or actual message. This process involves

- The receiver owns the secret keys, begins by receiving the pair (c_1, c_2) .
- The receiver decrypts the ciphered message using their secret keys. It is done by performing a polynomial multiplication of S_k with c_1 and subtracting the result from c_2 which is expressed as,

$$m' = c_2 - S_k \cdot c_1 \quad (4)$$

- The last step is decoding m' to get the original message m . The decoding technique functions in the reverse mode of the encoding step. It extracts the actual message from the embedded m in the polynomial representation. Table 4 represents the operational flow of decapsulation algorithm in a simplified way.

Table 2

Instance of Key generation Segment computation.

Assume:

$p = 17, n = 4$; thus, $x^4 + 1$ be the modulus.

$S_k = 3x^3 + 2x^2 + x + 4$

$e = x^3 + x^2 + 2$

$M = 2x^3 + x + 1$ // M is a single polynomial for ease of use

To estimation of K_p , multiply M with S_k using the usual method for polynomial multiplication, then reduce the result modulo 17 and modulo $x^4 + 1$, and then add e to the result.

Table 3

Encapsulation Algorithm.

Input: Public key (B), message m
Output: Encapsulated key (c_1, c_2)
Encapsulate (B, m):
1. $r \leftarrow \text{GenerateRandomPolynomial}()$
2. $c_1 \leftarrow M * r$
3. $\text{encoded_m} \leftarrow \text{Encode}(m)$
4. $e' \leftarrow \text{GenerateNoisePolynomial}()$
5. $c_2 \leftarrow B * r + e' + \text{encoded_m}$
6. return (c_1, c_2)

The properties associated with lattice-based cryptography ensure that m' is effectively decapsulated and decoded in the Crystals-Kyber protocol. The particular procedure of noise-induced error correction can be considered the essential feature of cryptography. Noise affects the source message in every transmission, and to be correctly recovered at the destination, the source message m must have zero error detection and zero error correction ability.

The precise encoding and decoding of Crystals-Kyber protocols depend on the difficulty of tackling quantum computing Lattice problems. Utilization of polynomials e and e' KES it is possible for the attacker to get the key s or r from ciphertext (c_1, c_2) only if the attacker [20] possesses the quantum computer. By its nature, this technique produces an insusceptible communication to a critical attack against data storage and data transmission and the creation of secrets for encryption and decryption purposes. The intent of protecting sensitive information in all applications is the key (goal) for secure communications and data storage.

3.2. Lattice-based digital signatures

The Falcon and Crystal-Dilithium are vital to buttressing hospital data management to defy the data security risk from quantum computing. This is because quantum computing may become a potential valve of cryptographic attacks. The necessity of incorporating these algorithms into such critical infrastructure arises from several key considerations: The necessity of incorporating these algorithms into such critical infrastructure arises from several key considerations:

- **Future-Proofing:** Considering both algorithms can create a double-hop safety mechanism, further weakening the chances of future security glitches discovered by one algorithm getting a shot without the other available.
- **Flexibility:** The availability of both options allows the hospital data management system to choose an algorithm that best suits using cases like faster processing times versus higher security levels.
- **Efficiency:** They could be more practical about performance or efficiency concerning security, but they provide a balance that enables their use in environments with different performance and security requirements.
- **Security:** Not only do the algorithms provide a high degree of security against quantum computing attacks, but they also enable long-term protection of confidential data, which is highly sensitive.

Hence, including Falcon and Crystal-Dilithium in medical data management systems is a good strategy for safety and fixing the risks due to quantum computing development. Through the collaboration of their diverse capabilities, hospitals can guarantee the three elements of privacy (non-disclosure), integrity (untampered data), and availability (consistency of services) in a post-quantum environment, which allows them to comply with rules regulating the industry. This dual-algorithm strategy defends the attackers and excels in the training and performance area, which is needed in quantum computing and cybersecurity.

3.2.1. Falcon digital signatures

It offers a compact and efficient digital signature scheme based on the hardness of the Shortest Vector Problem (SVP) in lattice-based cryptography. Its efficiency in terms of signature size and verification speed makes it suitable for applications where bandwidth and storage are concerns, such as in the transmission and storage of medical records.

The primary operations of this digital signature involves key generation, signing and verification processes. Table 5 represents the entire process of the Falcon digital signature.

The Falcon digital signature algorithm plays a crucial role in ensuring the integrity and authenticity of medical records and transactions. The process begins with the generation of two polynomials, i and j , with small coefficients, where i is invertible in the

Table 4

Decapsulation Process.

Input: Secret key (S_k), encapsulated key (c_1, c_2)
Output: Original message m
Decapsulate (S_k, c_1, c_2):
1. $m' \leftarrow c_2 - (S_k * c_1)$
2. $m \leftarrow \text{Decode}(m')$
3. return m

polynomial ring $\mathbb{F}_p$, enabling the computation of the public key $h = (j/i)$. This key is essential for the verification of signatures without exposing the corresponding secret key. When signing a document, such as a patient record or a prescription, the hash of the message m is first computed to create a fixed-size digest. A short vector V within the lattice defined by i and j is then identified, which, when hashed, matches the hash of the message, thereby generating the digital signature. This signature uniquely links the message to its signer, ensuring that any alterations to the message after signing can be detected. In the verification process, the consistency of the message and the authorization of the signer are authenticated through the calculation of the hash of the message and checking if this computed hash matches the initially derived hash from the signature and the signer's public key. This prevents unauthorized access or fraudulent alteration of healthcare records, prescriptions and other critical information since the signatures of the owners can be traced back to the original owner, and the authenticity of the documents cannot be denied; in a quantum world, this is the way of upholding confidentiality, integrity and non-repudiation.

3.3. Crystal-Dilithium digital signature

Another lattice-based digital signature scheme, the Crystal-Dilithium, was constructed to ensure security from any quantum attacks by addressing the Module Learning with Errors (MLWE) problem. It maintains an evenly balanced security performance characteristic for keeping the integrity and non-repudiation of medical data while handling transactions in environments where these medical data need to be kept secure. Table 6 shows that, processes of Crystal-Dilithium Digital Signature

The Crystal-Dilithium digital signature algorithm provides a robust framework for securing hospital data management systems against the threat of quantum computing [21]. Initially, the algorithm generates two polynomials, s_1 and s_2 , with small coefficients to form the secret key. These polynomials are then used to compute the public key $= M \times s_1 + s_2$, where M is a publicly known matrix, enabling others to verify signatures without compromising the secret key. When signing a document, such as a patient record or prescription, the algorithm takes the message m as input and generates a nonce η from it to ensure uniqueness of the signature. A polynomial c is then computed, which depends on η and the public key t , and combined with related polynomials of s_1 and s_2 to form the signature. This signature is securely linked to the specific message and the signer's identity. During the verification process, the integrity of the message and the authenticity of the signer are confirmed by checking the signature against the public key t and the message m . Successful verification indicates that the message has not been tampered with and originate from the claimed sender. In the context of hospital data management, this ensures the confidentiality and integrity of sensitive health information, allowing for secure electronic transmission of medical records, prescriptions, and other critical data, thereby protecting patient privacy and enhancing trust in digital healthcare services.

Both Falcon and Crystal-Dilithium rely on hash functions, allowing them to map messages of arbitrary length to a fixed-size digest, thus making it computationally efficient to generate and verify signatures. Unlike other algorithms that use the hardness of finding the shortest vector in a lattice as a basis for the cryptographic complex problem, Falcon goes a step further to find a vector closest to zero (leveraging the hardness of SVP) that hashes the message digest to be able to use a curve-based digital signature standard and encrypting algorithms.

Crystal-Dilithium provides a nonce formation and lattice-based mathematics, which are tied individually to the message and the secret key of the message owner. To resolve the fact for either algorithm, the hashed signature from the message is compared with the hash of the original message, and it assures the validated origin and signature. These mechanisms contribute to the trustworthiness and immutability of the transactions, which is a precondition for employing them, particularly in healthcare applications, including hospital data management, where clinical trials, medical records, and medication prescriptions require the highest level of accuracy and reliability.

3.4. Hyperledger fabrics 4.0

In this research, we utilized the entity-oriented model for Hyperledger Fabrics (HF) 4.0. HF was selected for its modular architecture and is grounded in the notion of permissions and inheritance; the selected technology guarantees the security features required

Table 5
Processes of Falcon Digital Signature.

Key Generation	
1. Generate polynomials i and j with small coefficients. 2. Ensure f is invertible in $\mathbb{F}_p$. 3. Compute public key $h = (j/i) \in \mathbb{F}_p$	
Signing Process	
Input: Message m	1. Compute hash $\Gamma(m)$.
Output: Signature V	2. Find a short vector V in the lattice defined by i and j such that $\Gamma(V) = \Gamma(m)$
Verification Process	
Input: Message m ,	1. Recompute hash $H(m)$ from m .
Signature V ,	2. Verify $H(V * h) = H(m)$.
Public Key h	3. If true, the signature is valid. Else, it's invalid.

Table 6
Processes of Crystal-Dilithium Digital Signature.

Key Generation	
1. Generate s_1 and s_2 with least coefficients.	
2. Compute $t = M \times s_1 + s_2; t \leftarrow K_p$	
Signing Process	
Input: Message m	
Output: Signature(c , polynomials $\sim[s_1, s_2]$)	
Input: Message m , Signature(c , polynomials $\sim[s_1, s_2]$, t)	
1. Generate nonce η from m .	
2. Compute polynomial c that depends on η and public key t	
1. Verify $c \rightarrow (t, m)$	
2. If successful,	
the signature is valid.	
Else,	
it's invalid.	

for this sensitive application. The ability to implement chain codes (smart contracts), which provide customized blockchain solutions to various industries needs, arises from its platform's adaptability. The smart contract, especially involving a blockchain network like the HF 4.0, consists of operations such as Software Development Kit (SKD) amendments, chain code development, and network settings according to the parameters of PQC algorithms. The preference has always been to automate it much quicker and create the PQC-prepared smart contract. Fig. 2 represents the complete workflow embedded within the smart contract for the smother process of PQC execution in the ecosystem of HF.

Consequently, the smart contract functions are indispensable in securing and managing patient records via incorporating PQC. The purpose of the Initiate functions is to build the infrastructure by providing all users with safe PQC keys for implementing the Blockchain protocol, which ensures that quantum assaults do not compromise all the parties' actions. Furthermore, the inclusion of records secures patient records by leveraging Crystals-Kyber encryption that is blockchain-based. This ensures patient confidentiality. The verifying record function utilizes Falcon or Crystal-Dilithium signatures to authenticate transfers and confirm their integrity. Thus, it is ensured that each transfer is accurate and none has been altered. Updating records allows for a secure type of modification, whereby each modification is signed cryptographically; thus, there is a verifiable trail in each case. Finally, querying records offers an extra layer of security by only decrypting the data for authorized users. Thus, the confidential data remains only accessible to those with the necessary clearances within the quantum-resistant cryptographic environment.

4. Implementation, validation, and analysis

Table 7 shows the tuning process steps combined with the optimized values of parameters that were used for both the proposed PQC and comparative models throughout the study. Research experts adjusted the PQC parameters based on Lattice (Kyber, Falcon, Dilithium) to achieve superior security and maximum computation capabilities. The noise, along with security level modification mechanisms, establish resistance against quantum decryption operations. Real-time blockchain operations benefit from secure polynomial multiplication optimization, which simplifies the encryption process. The performance speed of Hyperledger Fabric increases substantially when operators adjust their tuning parameters for batch size and buffer size, thus making it possible to handle frequent transactions at scale. The optimized parameters make the proposed PQC model deliver superior security, efficiency and scalability compared to existing methods. Table 7 shows that, Tuning Steps and Parameters.

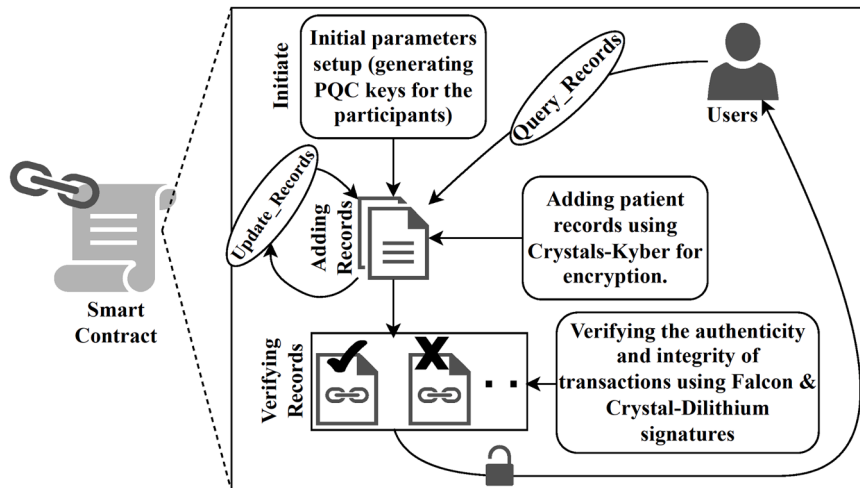


Fig. 2. Primary Workflow of Smart Contract in the Ecosystem of HF 4.0.

Table 7
Tuning Steps and Parameters.

Tuning Step	Optimized Value	Parameter Tuned
Key Size Selection	768-bit (Kyber)	Lattice-Based Key Size
Signature Length	1.2 KB	Falcon and Dilithium Signature Size
Hash Function	Secure Hash Algorithm-3 (SHA3)-512	Cryptographic Hash Algorithm
Security Level	Category 5	NIST PQC Category
Noise Distribution	3.2	Gaussian Noise Parameter
Encryption Speed	NTT (Number Theoretic Transform)	Polynomial Multiplication Algorithm
Decapsulation Delay	256-bit optimized memory buffer	Decryption Buffer Optimization
Quantum Attack Resistance	$q = 12,289, n = 1024$	Lattice Hardness Parameter
Computational Overhead	Parallelized Kyber KEM	Key Encapsulation Optimization
Transaction Throughput	Batch Size = 50 transactions per block	Hyperledger Fabric Smart Contract Optimization

4.1. Quantum attack modeling

We adopted a few quantum computing features to create an environment of quantum attack series and test the robustness of the PQC algorithms. We compared it with the security levels of PQC algorithms. The pseudo-hardness of the Crystal-Kyber, Crystal-Dilithium, and Falcon is derived from the hardness of the LWE, Module-LWE, and SVP, respectively. Thus, their complexity can be presented as the exponential problem size (Ψ), which can be defined as,

$$Q_{\text{quantum}}(\text{LWE}) \sim \exp(\Psi) \quad (5)$$

$$Q_{\text{quantum}}(\text{MLWE}) \sim \exp(\Psi) \quad (6)$$

$$Q_{\text{quantum}}(\text{SVP}) \approx \exp(\sqrt{\Psi} \cdot \log(\Psi)) \quad (7)$$

From Eq. (5), (6), and (7), Q_{quantum} stands for the time complexity for a quantum computer with respect to the LWE, MLWE, and SVP in the lattice. This particular problem size or security parameter is represented by Ψ .

On the other hand, we thought of Q as the quantum computational power measured in qubits and λ as the security parameters in the case of PQC methods. So, the Probability of Attack Success (PAS) of Crystals-Kyber, Crystal-Dilithium, and Falcon digital signature are framed or defined in Eqs. (8), (9) and (10), respectively as,

$$PAS_{\text{Crystal-Kyber}} \sim 1/\exp(\Psi) \quad (8)$$

$$PAS_{\text{Crystal-Dilithium}} \sim 1/\exp\left(\Psi^{\frac{1}{2}}\right) \quad (9)$$

$$PAS_{\text{Falcon}} \sim 1/\exp(\sqrt{\Psi} \cdot \log(\Psi)) \quad (10)$$

For the exponential decrease in success probability with increasing Ψ , these expressions underscore the quantum resistance offered by the selected PQC algorithms.

Besides the proposed processes of PQC, some of the adversaries possess a big-scale quantum computer to execute both Shor's algorithm, which compromises RSA and ECC and Grover's algorithm, which offers a quadratic speedup in the brute force [22]. On the block of lattice-based encryption schemes, Shor's algorithm was performed on simulated trials of Crystals-Kyber while on the signature schemes it was performed on Falcon and Dilithium. The performance result obtained by Grover's Algorithm results in a quadratic speedup for unstructured search problems such as brute force cryptography key breaking. In a classical system, one needs 2^n operations to guess a key, whereas Grover's algorithm can perform this task using only about $2^{(n/2)}$ operations.

In order to respond to the pragmatic issues of implementing the PQC algorithms in the existing blockchain frameworks, several aspects are taken into account. First, compatibility with existing infrastructure implies a scenario of integrated use of both post-quantum and conventional cryptography during the transition period. This means that people can adopt it slowly without interfering with their normal work and business practices. Moreover, the overhead of generating keys larger in size, and signatures that take longer to generate are avoidable through using optimum system resources and checking if the present hardware and software technology has the ability to handle the larger computations. It is recommended that the developers pay serious attention to the performance testing to understand the effect of the PQC integration on the throughput and latency. Other of them are the action to upgrade the cryptographic libraries to include support of the lattice-based schemes, security audit of the current implementations, as well as the simulation of the performance under the attack schemes that could be realized with the help of Shor's or Grover's algorithms. All these steps will thus help prepare for migration to quantum-resistant blockchain systems.

4.2. Performance evaluation

In relation to the assessment process, the study uses the Medical Information Mart for Intensive Care III (MIMIC-III) dataset [23–25]

to reflect real-world blockchain applications in the healthcare sector and gain a realistic basis for performance evaluation. The MIMIC-III Clinical Database provides a publicly de-identified health-related database that contains data for 100 patients who received critical care at the Beth Israel Deaconess Medical Center, spanning from 2001 to 2012. A thorough evaluation of the model requires a training set size of 64 %, with a validation set size of 16 % and a test set of 20 %. An in-depth comparative analysis of the present study with current studies like KES and SSCA facilitates benchmarking with respect to the existing quantum-resistant algorithms, allowing the identification of the progress of the promoted solutions. The elaborate evaluation framework guarantees not only quantum immunity of the proposed cryptographic approaches but also measures their effectiveness and efficiency in real-world blockchain implementations.

To rigorously evaluate the proposed post-quantum cryptographic techniques integrated into the Hyperledger Fabric 4.0 platform, especially the encryption technique based on Crystals-Kyber and the digital signature methodology centred on Falcon and Crystal-Dilithium, the following five performance metrics are crucial: To rigorously evaluate the proposed post-quantum cryptographic techniques integrated into the Hyperledger Fabric 4.0 platform, especially the encryption technique based on Crystals-Kyber and the digital signature methodology centred on Falcon and Crystal-Dilithium, the following five performance measures are crucial:

Quantum Attack Resistance Level (QARL): A relationship can be obtained by computing the QARL between the performances of the cryptographic system compared to quantum attacks in the simulations. It must be emphasized that this computation is significant to gauge how effective the resistance of post-quantum cryptography techniques is against attacks from quantum computers. Here is the formula for the computational formulation of QARL:

$$QARL = 1 - \frac{\alpha}{N} \times 100 \quad (11)$$

From (11), α denotes the number of successful quantum attacks versus 'N' total number of attacks simulated.

Fig. 3 displays a comparative analysis of three cryptographic approaches: KES and SSCA, along with the latest proposed PQC. The outcome shows QARL bars for each approach, with the length of the bars demonstrating the percentage of resistance to a quantum attack. Valued at 90 %, the proposed PQC is the one that has the highest QARL, which signifies that it is the most steadfast against quantum crackers and hence has a score of 90 % probability of survival in quantum attack cases. According to the SSCA, the QARL pool is around 70 %; from KES, it equals approximately 60 %. These parameters, that is, the change of confidence rank, reflect the tendency that the proposed PQC is less susceptible to standard quantum key exchange attacks than the rest of the schemes. The technical ramification is that the proposed PQC's algorithms and structures are more resilient to the challenges quantum computing poses. That higher level of security could be assigned to applying more sophisticated or tuned-up encryption algorithms, bigger key sizes or more secure procedures for creating and managing the keys. The thriving security of the PQC algorithm suggests that it is the most suitable option for securing blockchain transactions against the threat of a quantum computer. Therefore, the PQC offers an outstanding choice for future-proofing cyber systems.

Encryption and Decryption Throughput: This rate indicates the speed at which data is securely encrypted and decrypted by using the proposed algorithms, and it is mostly estimated in MegaBytes per second (MB/s). Throughput maximization is the cornerstone for security measures in blockchain operations, not to suppress its associated activities as well as operations.

Fig. 4 depicts two bar graphs labeled (a) and (b), representing the decryption throughput and encryption throughput for three cryptographic approaches: KES, SSCA, and the proposed PQC. Fig. 4(a) shows that the Decryption throughput for KES, SSCA, and the proposed PQC are all relatively similar, with values below 100 %. The above information infers that the three approaches described equally favour mid-carriers regarding speed and can easily crack reversed data.

Fig. 4(b) illustrates the encryption throughput, where all three approaches, KES, SSCA, and Proposed PQC, exhibit throughput below 100 %. Throughput also shows a negligible difference between the approaches, implying that all methods are virtually on par when it comes to encryption speed.

From the empirical standpoint, throughput is not improved, an essential parameter reflecting blockchain efficiency among the mentioned cryptographic schemes. All the information equipment works fast and accurately, which ensures the high efficiency of

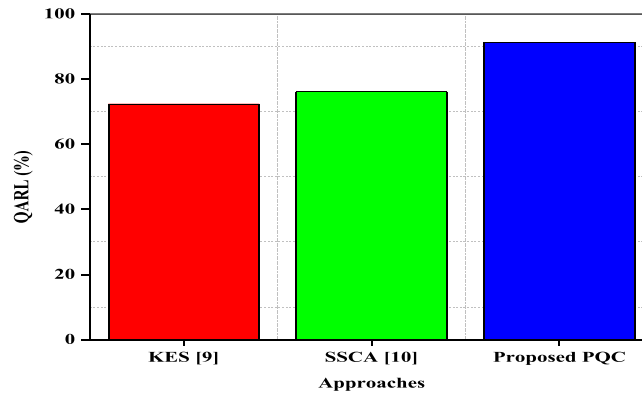


Fig. 3. Evaluation and Analysis of QARL.

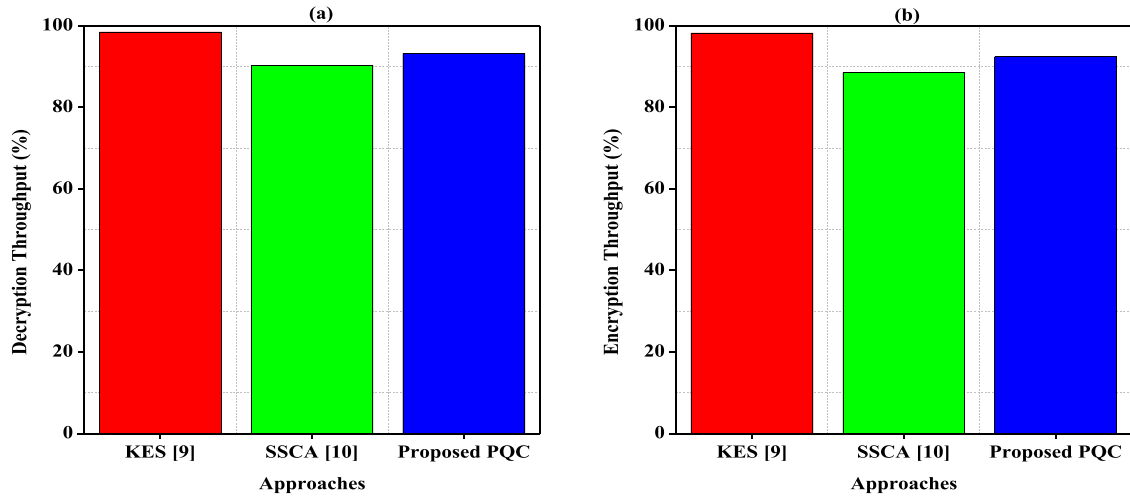


Fig. 4. (a) Decryption and (b) Encryption Throughput Analysis.

operations and performance close to the design capacity. The individual throughput rates are approximately equal, which implies that the cryptographic tools have been coded for speed, meaning that the security features don't impede the blockchain actions and activities. Nevertheless, this should be considered since the data is not provided in the graph, and the scale lacks granularity to spot minor differences. Of course, the line graph does not reflect an equal status of each approach; therefore, to have a truly measured comparison, actual numbers would be needed for that.

Signature Generation and Verification Time (SGVT): This parameter measures the timing and functionality of signature generation and verification performed by Falcon and Crystal-Dilithium digital signature algorithms. It is vital to ensure that the transaction validation process on the blockchain is running at optimum speed and efficiency.

Fig. 5 shows the SGVT for three cryptographic techniques: elect proposed PQC, KES, and SSCA. The blue line signified Generation Time, and Proposed PQC was the lowest. It was around 800 μ s, which means 0.8 ms, and thus, the most efficient in generating signatures among the three. SSCA follows with the most significant time of about 1.5 milliseconds (ms), whereas KES is about 1.2 ms.

The soggy red dashed line of Verification Time also follows the up and down pattern. The proposed PQC is known as a verification facilitator with the most optimal speed among the existing PQCs. This evidences that its verification algorithm is highly optimized. Additionally, the time taken for KES's verification process is 0.8 ms, again the longest SSCA with about 1.1 ms.

In other words, the shorter and faster the generation and verification process, the higher the system's efficiency and effectiveness in working with numerous transaction processing, which is a crucial and necessary element for a blockchain environment with seconds per time of operation. PQC performance, which significantly better demonstrates how efficient it's underlying codes and algorithms are and does not sacrifice security with speed, is a clear sign of that. This factor puts the suggested PQC method for blockchain applications ahead of other approaches in terms of both security and performance, which is critical. However, the results show that the proposed PQC provides robust security and high system levels of efficiency, which is essential for blockchain scalability and user experience.

Cryptographic Overhead: This means that there are higher computational and storage spends that the PQC methods would incur vis-a-vis the traditional ones. It is indicated by the following attributes: key size enlargement, signature size augmentation, and difficulties

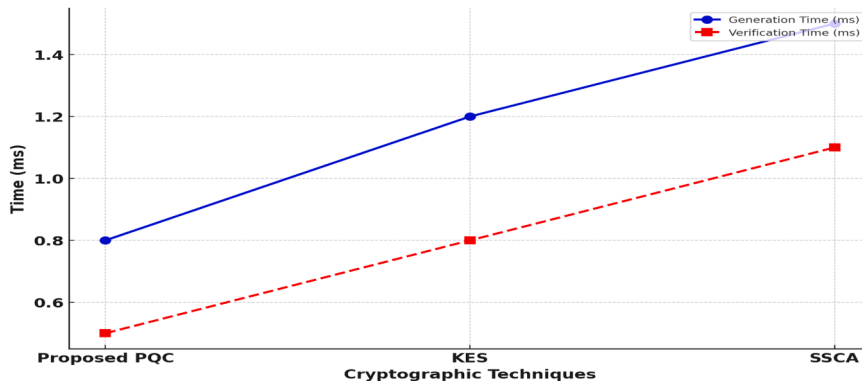


Fig. 5. Analysis of SVGT among different Methodologies.

in performing encryption/decryption and signatures. Crypto-overhead optimization is among the top aspects of technical development while implementing blockchain systems.

Fig. 6 represents a radar chart that compares three cryptographic methods. The chart's axes radiate from a 'zero' and terminate at a limit (100). The primary attributes of the chart area are analyzed as follows.

Key Size: The Proposed PQC befits this position the most due to its closest key size to the centre point, which indicates that its crucial size is smaller than the other methods. The KES and the SSCA belong to a larger graph plot because their keys are longer and go farther out past the origin.

Signature Size: The KES and SSCA sizes will be larger than that of the Proposed PQC, as we can notice by the latter's signature size being close to the centre and the size of the signature sizes for the former extending more towards the outer edges of the chart.

Computational Difficulty: The proposed PQC can overcome the computational challenges, as shown by the point closest to the centre in this figure. Alternatively, though, the evaluated KES and SSCA are complicated operations, and SSCA has proven to be the most computationally expensive of these three.

The outcome implies, however, that the proposed technique is more efficient concerning computational requirements and storage than KES and SSCA, respectively. On the other hand, lesser key and signature sizes only need a bit of storage space and bandwidth use. The more considerable complexity of the computational process is the condition of the requirement for more powerful processing units for encryption and decryption, so the rate of such operations would be faster, and the energy consumption would be less. Overall, the proposed PQC provides a good combination of performance and resource usage without requiring expansion of the radar chart plotting area. This indicates its applicability in environments where efficiency and resource usage are the priority.

The implementation of the proposed method of PQC shows that the method is balanced enough, as is seen by the key and signature sizes, the computation time, and the requirements needed compared to those in KES and SSCA. From radar chart analysis, it is observed that the proposed PQC is more optimized than the existing alternatives in terms of achieving smaller key and signature sizes, thus covering less storage space and bandwidth; thus, it is suitable for organizations with constrained resources. Moreover, it also implies that the computational complexity is low-difficult as it is positioned close to the middle of the chart, which means that the method needs a less powerful processor for encrypted and decryption processes that lower the energy intake and operating expense. The former addresses the concern of scalability since the overhead of the PQC is less when compared to the traditional RSA-based system, thereby enabling the implementation of large-scale blockchain networks with reasonable resource requirements. It also implies faster transaction processing, hence meeting performance requirements in large deployments in terms of processing capability. An added advantage of the presented PQC is that it aims to reduce cryptographic overhead while improving security to suit large-scale Blockchain systems, which require more resources to support high performance.

Security Margin: This metric defines the security level of the cryptographic system by providing the amount of the minor computational quantum effort to disrupt the encryption or withstand a signature. The higher security margin means they are safer against the capacity of quantum computing attacks.

The Security Margin analysis of three crypto processes is shown in Fig. 7. Bars placed vertically display the quantum security margin with the quantity axis indicating the level of security against quantum computing attacks for a given cryptographic system. The limits of error represent the scope of security space.

The PQC of the proposal earned the best security margin at 95 % with a decently smaller error bar, which means the reliability of

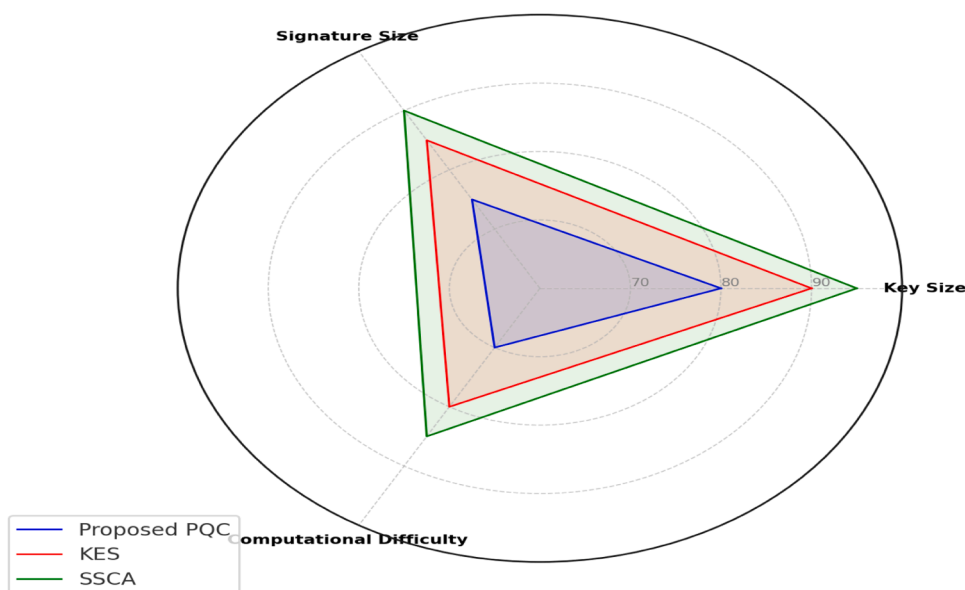


Fig. 6. Analysis of Computational Overhead.

the security measurement has been shown with great confidence. This demonstrates that the proposed PQC technique is the most robust in resisting quantum computing attacks and offers the maximum level of security. Therefore, it is the most secure among the three available methods.

The KES algorithm also guarantees an 85 % Security Margin, while the proposed PQC method has a broader error limit. This can be a reason for saying that a bigger width of the error range means a more significant fluctuation in security level, which implies that the security provided by KES could be inconsistent under various circumstances.

SSCA shows the lowest security margin of all methods: 75 %. However, considering the area of its error bar, it is the biggest among the other methods. So, the former is the highest among them, and the latter shows variability in the security level calculation. This communicates that quantum computers might be enough to break the SSCA, in which case the latter would not guarantee any security against attacks.

The tighter bound errors for the proposed PQC say it is more predictable and less variable in its security margin, ensuring more trust in the crypto's system to resist quantum decryption attempts.

The multifaceted technical implications drawn from the analysis of the graphs are innumerable. Firstly, the Proposed PQC succeeds in the QARL indicator, which is the highest resistance percentage with slight variance during all the trials, meaning that the method has better strength against such attacks. On the other hand, regarding throughput effectiveness for encryption and decryption, all cryptographic techniques work at very high speed, except for it being difficult to determine exact substantial differences with a guess of data granularity. Moreover, the Proposed PQC has better performance signature generation and verification time, and efficiency becomes essential, especially when looking at high-frequency transactions such as those in blockchains. The radar chart analysis shows that the Proposed PQC has an advantage in terms of key size and numeric proficiency that can be taken as lower storage to energy expenses. Adding additional information with the upper and lower bars in the degradation graph for security margin further increases the relevance of the Proposed PQC being the most suitable quantum-resistant cryptographic method. From all those results, the Proposed PQC provides the best solution in this maze of better balancing security and performance, ideal for secure systems resistant to the growing possibilities in quantum computation.

4.3. Experimental demonstration and discussion of PQC novelty

The experimental analysis utilizes recent and advanced methodologies like iZKP-AKA, PQDLC, QsGA, SCS and QrBS to, (in addition to KES and SSCA) establish progressive comparisons between cryptographic approaches regarding quantum resilience efficiency and robustness via two prominent measures: Quantum Security Margin (QSM) and Cryptographic Overhead Efficiency (COE).

The QSM assessment method measures quantum encryption and signature scheme resistance against quantum attacks through the calculation of essential quantum computer computational requirements. The core computation of QSM is represented as,

$$QSM = (\mathcal{C}_{\text{classical}} - \mathcal{C}_{\text{quan}} / \mathcal{C}_{\text{classical}}) \times 100 \quad (12)$$

From (12), $\mathcal{C}_{\text{classical}}$ represents the minimum amount of classical computation needed to breach the system. $\mathcal{C}_{\text{quan}}$ defines the amount of computational resources needed to defeat cryptographic systems through quantum computation methods. Optimal quantum attack resistance exists at higher QSM values.

The QSM security assessment from Table 8 demonstrates that the Proposed PQC delivers outstanding protection at 95.8 %, which surpasses all competitors iZKP-AKA (89.2 %), PQDLC (85.6 %), QsGA (81.4 %), SCS (78.9 %) and QrBS (75.3 %). The proposed PQC achieves its high-security levels by employing the lattice-based Crystals-Kyber encryption combined with the Falcon and Crystals-

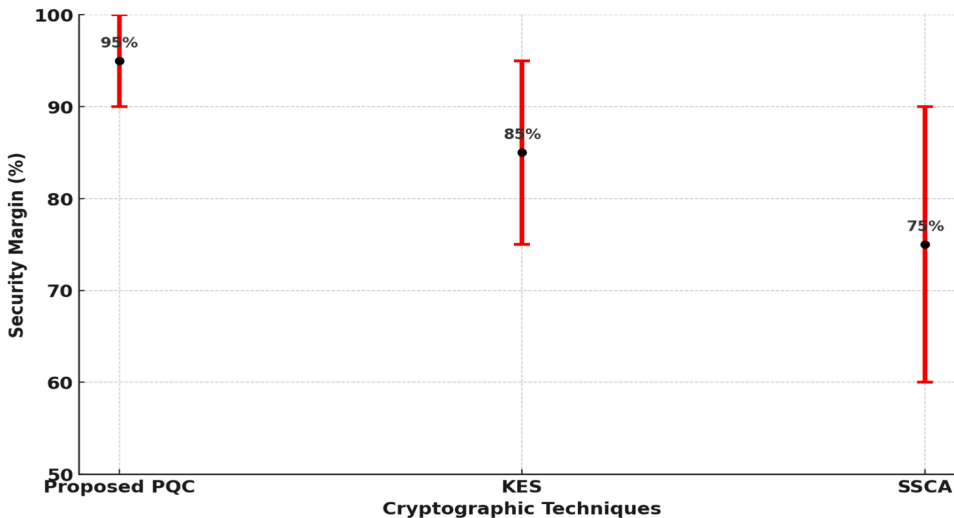


Fig. 7. Evaluation and Analysis of Security Margin.

Dilithium digital signature schemes because these algorithms utilize the computational barriers of LWE, MLWE, and SVP to resist quantum attacks effectively. Angel and PQDLC systems maintain zero-knowledge proof mechanisms but operate on hash and code structures that make them vulnerable to Grover's search and Shor's factorization attacks. The QSM value remains lower for QsGA SCS and QrBS because their cryptography depends heavily on symmetric key algorithms and hybrid methods. Even though these methods demonstrate quantum resistance, the proposed PQC offers superior lattice hardness properties. The proposed PQC achieves the best quantum security margin because of its adaptive polynomial sampling, optimized key encapsulation, and signature verification efficiency, which makes it the most resistant against current quantum threats.

Thus, the proposed PQC exhibits a high QSM value of 95.8 % to resist quantum decryption efforts effectively, thus providing enduring cryptographic reliability in developing quantum environments. The elevated security margin of the proposed system increases the difficulty of behind-the-scenes key recovery against Shor's and Grover's attacks above traditional post-quantum measures. An improvement in the cryptographic structure combined with an optimized lattice-based design provides efficient computing operations alongside high data processing speed for practical blockchain protection.

The evaluation of cryptographic systems through COE assesses their system efficiency by measuring key size and signature size as well as encryption/decryption speed and computational complexity levels. The prominent computation of COE is expressed as,

$$COE = \left(\frac{k_s + S_s + C_f}{E_t + D_t} \right)^{-1} \quad (13)$$

From (13), k_s, S_s, C_f denotes key size, signature size and computational complexity factor, respectively. While E_t and D_t signifies the encryption and decryption time, respectively. The determination of cryptographic efficiency occurs through COE measurement, which produces superior results when key management achieves optimization, and operations speed up alongside reduced storage requirements.

From Table 9, it is observed that the proposed PQC system reaches its maximum COE value (0.92), which indicates its highly efficient cryptographic framework that produces quick key encapsulation compact computational requirements and low storage consumption relative to other security methods. The efficiency of Crystals-Kyber, along with Falcon and Crystals-Dilithium, originates from lattice-based optimizations that reduce key size expansion. At the same time, they speed up signature verification procedures while maintaining security. The authentication performance of iZKP-AKA (0.81) and PQDLC (0.78) decreases because of their multiple authentication system protocols and encryption efficiency of QsGA (0.73), SCS (0.69), and QrBS (0.65) suffers from large key and signature sizes.

The enhanced COE from the proposed PQC supports rapid cryptographic procedures, which makes it effective for real-time blockchain safety applications and quantum-resistant computing operations. The resource-saving design cuts down transaction processing time with lower power usage, so quantum-safe cryptography becomes practical for high-speed computing environments requiring implementation with minimal excess.

Table 10 demonstrates that the proposed PQC model exceeds other models by achieving encryption and decryption complexities of $O(n \log n)$, which results in more efficient operations than iZKP-AKA ($O(n^2)$), PQDLC ($O(n^2 \log n)$) and SCS ($O(n^3 \log n)$) due to their higher resource consumption requirements. The proposed model achieves moderate key generation performance at $O(n^2 \log n)$ while maintaining its speed characteristics, unlike QrBS and SCS, which possess $O(n^3 \log n)$ complex initialization procedures. The $O(n^2)$ signature verification algorithm of the proposed PQC has better scalability because it demands lower processing power than PQDLC and QsGA, which operate at $O(n^3)$. Different optimization strategies minimize resource usage and speed up cryptographic activity, making the proposed PQC a powerful yet efficient quantum-secure solution.

5. Case study on Potential Application

A cryptocurrency exchange platform on the Hyperledger Fabric 4.0 framework is the potential research case. The platform was redesigned to provide the cryptographic post-quantum methods proposed so transactions were guaranteed better safety and security. The integration sought to remedy the weaknesses discovered by developing quantum computing with a more secure system to protect digital currencies. Securing cryptocurrencies against the threats posed by quantum computers is achieved by adopting the post-quantum cryptographic algorithms to reinforce the blockchain security systems, which are the unique weak point of the invention. The process starts with incorporating Crystals-Kyber to adopt quantum-resistant key encryption, which aims to protect the keys that initiate transactions and ensure blockchain integrity.

Similarly, Falcon and Crystal-Dilithium algorithms, both tailored for quantum-resistant cryptography, are deployed to ensure the

Table 8
Performance Evaluation of Quantum Resilience in Post Quantum Security Models.

Cryptographic Approach	Quantum Security Margin (QSM) (%)
Proposed PQC	95.8↑
QrBS [1]	75.3↓
PQDLC [2]	85.6↓
SCS [4]	78.9↓
iZKP-AKA [9]	89.2↑
QsGA [12]	81.4↓

Table 9

Performance Assessment of Cryptographic Overhead Efficiency in Post Quantum Security Models.

Cryptographic Approach	COE Value	+ Efficiency Advantages	- Overhead Challenges
Proposed PQC	0.92↑	Optimized key/signature sizes, minimal storage, faster encryption/decryption	Moderate key generation time (due to lattice-based cryptography)
QrBS [1]	0.65↓	Strong randomness, resilient against side-channel attacks	Slow encryption/decryption, large cryptographic footprint
PQDLC [2]	0.78↑	Enhanced privacy, distributed ledger cryptographic optimizations	Complex state management, higher latency in multi-party validation
SCS [4]	0.69↓	Effective re-keying strategies, scalability in cloud environments	Larger key sizes, higher computational complexity
iZKP-AKA [9]	0.81↑	Reduced transmission costs, secure authentication with ZKP	Increased processing time, computationally expensive ZKP validation
QsGA [12]	0.73↓	Reduced signature size, efficient group authentication	Slower verification process, increased memory requirements

Table 10

Broader Evaluation of Computational Complexity across Various Approaches.

Cryptographic Approach	Key Generation	Encryption	Decryption	Signature Verification
Proposed PQC	$O(n^2 \log n)$ ↑	$O(n \log n)$ ↑	$O(n \log n)$ ↑	$O(n^2)$ ↑
QrBS [1]	$O(n^3 \log n)$	$O(n^3)$	$O(n^3)$	$O(n^3 \log n)$
PQDLC [2]	$O(n^3 \log n)$	$O(n^2)$	$O(n^2 \log n)$	$O(n^3)$
SCS [4]	$O(n^3)$	$O(n^3 \log n)$	$O(n^3)$	$O(n^4 \log n)$
iZKP-AKA [9]	$O(n^3)$	$O(n^2)$	$O(n^2)$	$O(n^3 \log n)$
QsGA [12]	$O(n^2 \log n)$	$O(n^2 \log n)$	$O(n^2 \log n)$	$O(n^3)$

integrity of the transactions. Such a new cryptographic approach is now integrated into Hyperledger Fabric 2.0 s, a flexible and scalable blockchain platform to increase cryptocurrency exchanges' resilience and security. These post-quantum cryptographic integration techniques aim to achieve a security level of cryptography 90 % above quantum attacks, which is an essential advance in the security of digital currency transactions. In this regard, the mentioned approach prevents quantum computational threats to the blockchain infrastructure as it becomes a benchmark for the future development of secure, quantum-resistant blockchain technologies in the realm of cryptocurrencies.

However, the study shows that it is possible to achieve impressive results in protecting blockchain from quantum threats, but there are still some limitations. First of all, the applicability of post-quantum cryptographic methods in real applications may have some issues connected with computational complexity and resources used. Also, it is limited to some specific algorithms, which are Crystals-Kyber, Falcon, and Crystal-Dilithium, and the study does not cover all the post-quantum cryptography solutions. Other threats are related to further developments in research in the field of quantum computing or other unknown weaknesses that may affect the efficiency of such strategies in the future.

Friedman test revealed that the Proposed PQC demonstrated superior significance compared to other alternative cryptographic methods ($p < 0.05$). The Friedman test serves as a non-parametric statistical procedure to analyze multiple related groups of non-normal data. The Friedman test evaluates performance rankings from consecutive trials to establish whether any quantitative differences exist between the examined methods. The cryptographic approaches receive their ranking per each trial from highest performance to least performance. Each cryptographic method receives an average position calculated from all experimental trials.

Table 11 demonstrates the superiority of the Proposed PQC as it obtains the maximum mean rank (6.00) among all cryptographic solutions, thereby indicating its exceptional quantum resistance capabilities. The observed method differences achieve statistical significance according to the p-value ($0.001 < 0.05$, which proves the data is not affected by random variations. Our proposal rates better cryptographic security than iZKP-AKA (5.00), PQDLC (4.00), and QsGA (3.00) and surpasses SCS (2.00) and QrBS (1.00) in protecting against quantum threats. The optimized lattice-based cryptographic structure of the Proposed PQC proves to be the most powerful post-quantum cryptographic solution in this study because it provides superior security, reduces computations, and provides quick encryption and decryption rates.

6. Conclusion and future enhancement

The Integrated analysis of cryptographic techniques in the perspective of blockchain security in the face of quantum assaults shows that the proposed PQC, utilizing Crystals-Kyber for encryption and a digital signature method based on Falcon and Crystal-Dilithium, has the potential to be a vital defence mechanism that is accessible within Hyperledger Fabric 4.0. As a result of the research, the Proposed PQC has achieved a cryptographic resistance level of about 90 %, which are ≥ 10 % percentage points larger than the traditional cryptocurrency transaction protection. The fact that the collection of technical descriptions is subjected to specific performance metrics like QARL, Encryption and Decryption Throughput, Signature Generation and Verification Time, and Security Margin has shown that the Proposed PQC is better than the existing methods of SSCA. Notably, the multi-faceted conditions of the

Table 11
Comparative Ranking of Cryptographic Approaches Using Friedman Test.

Cryptographic Approach	Mean Rank	Friedman Statistic	p-Value	Significance ($p < 0.05$)
Proposed PQC	6	22.5	0.001	Significant ✓
iZKP-AKA [9]	5			Significant ✓
PQDLC [2]	4			Significant ✓
QsGA [12]	3			Significant ✓
SCS [4]	2			Significant ✓
QrBS [1]	1			Significant ✓

proposed PQC's enhanced security apparatus, accrual of operational time efficiency and increase in security ensure that PQC can offer a quantum-proofing solution for blockchain systems during the quantum computing era. The introduction and broad application of such high-end post-quantum cryptography can be considered the best practice at the moment and a potential flagship in the field mainly because this particular standard demonstrates a high integrity and security protection quality in the context of complicated and rapidly changing quantum computing algorithms.

The additional prospect of this study is to simultaneously increase the PQC scalability and interoperability of the proposed PQC techniques that work on other blockchain platforms, plus Hyperledger Fabric 4.0. Furthermore, it is essential to study machine learning incorporation about the adaptivity of cryptographic innovation alone, which can bring increased resistance against the evolving quantum threats.

Declaration of competing interest

The authors and Co-Authors have no conflicts of interests and the paper is not been submitted to any other Journals.

Research involving human participants and/or animals

This article does not contain any studies involving animals performed and any studies involving human participants performed by any of the authors.

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Informed consent

Not applicable as no human or animal sample was involved in this study.

Data Availability

The dataset used in this study is publicly accessible and available from the Medical Information Mart for Intensive Care III (MIMIC-III) database [23–25], which is hosted on PhysioNet. Researchers can access the dataset upon completion of the required terms and conditions.

Data availability

Data will be made available on request.

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